

Endpoint L^1 Vertical Inequalities on Nilpotent Groups

Verified Theorems, Lorentz Propagation, and the Cartan Barrier

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Abstract

Ryoo asked whether every nonabelian simply connected nilpotent Lie group admits a finite scale-summability exponent in the endpoint L^1 vertical-versus-horizontal inequality along a last-layer direction [8]. This note records the strongest conclusions obtained in a sustained attempt to answer that question. The complete step-two case is proved with exponent 4, in fact for arbitrary bracket-generating left-invariant Carnot–Carathéodory structures on step-two groups. A universal mixed-bracket theorem is proved in the full Lorentz scale: regularity and scale summability propagate from a direction W to $[X, W]$, with the exact Carnot-weight transformation. The weak endpoint version is particularly useful because a finite weak scale exponent, together with the deterministic path bound, automatically gives every larger strong exponent. Consequently, the whole step-three problem—including arbitrary bracket-generating left-invariant horizontal structures—is equivalent to a single finite weak-type estimate for one top-layer direction on the five-dimensional Cartan group. A still more structured weak estimate for its noncentral second-layer direction would imply the strong step-three theorem for every exponent greater than 6. More generally, one weak rank-two seed on the free group $F_{2,s}$ yields the full strong theorem in step s above an explicit exponent. These reductions also give unconditional higher-step examples, including exponent $2s$ for the top direction of $UT_{s+1}(\mathbb{R})$.

No proof or counterexample for the remaining Cartan weak estimate is obtained here. The final sections explain why quotient lifting and plain Heisenberg slicing do not close, and isolate the missing statement as a finite-index kinetic/Carleson cancellation estimate. The results in this note are intended as a rigorous research record and have not been peer reviewed.

Status. The general conjecture is not proved in this note. What is proved is: (i) the complete step-two theorem; (ii) strong and weak Lorentz mixed-bracket propagation; (iii) permanence under subgroups and nilpotent quotients; (iv) an exact weak-type reduction of all step-three nilpotent groups to the Cartan group; (v) a rank-two weak seed theorem in arbitrary step; and (vi) unconditional higher-step structured examples. The sole universal step-three target is the weak Cartan estimate in equation (32) below. This packet supersedes the previous step-two-only packet, which is preserved in `history/previous_step_two_packet_2026_06_19/`.

Source-question screenshot

The source paper is Seung-Yeon Ryoo, *Quantitative nonembeddability of groups of polynomial growth into uniformly convex spaces*, arXiv:2207.11305. The targeted open endpoint problem is Questions 38 and 42 / Conjecture 43, asking for finite scale-summability exponents in endpoint L^1 vertical-versus-horizontal inequalities.

Question 38. Let G be a nonabelian simply connected nilpotent Lie group of nilpotency step s , and let $v \in \underbrace{[G, [G, \dots, G]]}_{s \text{ times}}$ be normalized so that $d_G(v, e_G) = 1$. For which exponents $q \geq 1$ does every smooth and compactly supported $f : G \rightarrow \mathbb{R}$ satisfy the following?

$$\left(\int_0^\infty \left(\int_G |f(h \exp(tv)) - f(h)| d\mu(h) \right)^q \frac{dt}{t^{1+q/s}} \right)^{1/q} \lesssim_{G,q} \int_G \|\nabla f(h)\|_{\ell_2^*} d\mu(h). \quad (26)$$

This tensorizes easily into an inequality for functions $f : G \rightarrow L^1$. Similarly, we may ask the following question.

Question 39. Let Γ be a not virtually abelian finitely generated group of polynomial growth. Choose v_Γ and $s > 0$ as in Theorem 7. For which exponents $q \geq 1$ do there exist $c = c(\Gamma) \in \mathbb{N}$ such that for any $n \in \mathbb{N}$ and $f : \Gamma \rightarrow L^1$ we have the following?

$$\left(\sum_{k=1}^{n^s} \frac{1}{k^{1+q/s}} \sum_{x \in B_n^\Gamma} \|f(xv_\Gamma^k) - f(x)\|_{L^1}^q \right)^{1/q} \lesssim_{\Gamma,q} \sum_{x \in B_{cn}^\Gamma} \sum_{a \in S} \|f(xa) - f(x)\|_{L^1}.$$

By the discretization argument of Section 3, an exponent q that answers Question 38 positively also answers Question 39 positively, when G is chosen as the Malcev completion of the torsion subgroup quotient of a finite index nilpotent subgroup of Γ .

Because the case $q = \infty$ formally holds for Question 38 (see Remark 29), by Hölder's inequality the set of q that satisfies Question 38 is either of the form $(q_{G,v}, \infty)$ or $[q_{G,v}, \infty)$ for some $q_{G,v} \in [1, \infty]$ depending on G . We make the following formal definition.

Definition 40. Given a nonabelian nilpotent Lie group G , let $v \in \underbrace{[\mathfrak{g}, [\mathfrak{g}, \dots, \mathfrak{g}]]}_{s \text{ times}}$ with $d_G(\exp(v), e_G) = 1$. Let $q_{G,v}$ denote the infimum of those $q \geq 1$ such that (26) holds for every smooth and compactly supported function $f : G \rightarrow \mathbb{R}$. Let q_G be the infimum of $q_{G,v}$ over all choices of v as above.

For $G = \mathbb{H}^3$, there is only one choice of v up to a sign, and we have $q_{\mathbb{H}^3} = 4$ with the exponent range satisfying (26) being $[4, \infty)$ [NY22]. For $G = \mathbb{H}^{2k+1}$ with $k \geq 2$, there is again only one choice of v up to a sign, and $q_{\mathbb{H}^{2k+1}} = 2$ with the exponent range satisfying (26) being $[2, \infty)$ [NY18].

For a general nilpotent Lie group G , we wish to know whether there are finite exponents q satisfying (26), since then not only would we reprove [EBGLD⁺23], but we would also have analogous quantitative nonembeddability statements: whenever q satisfies (26), then, similarly to Theorem 1, if G is Riemannian, we would have

$$c_1(B_r) \gtrsim_{G,q} (\log r)^{1/q}, \quad r \geq 2,$$

and similarly to Theorem 19, if G is Riemannian or sub-Riemannian and if N_{r_1, r_2} is an r_2 -covering of B_{r_1} , where $r_1, r_2 > 0$ with $r_1 \geq 2r_2$, and with the extra condition that $r_1, r_2 \geq 1$ if $v \in \text{span}\{X_1, \dots, X_k\}$, then

$$c_1(N_{r_1, r_2}) \gtrsim_{G,q} (\log(r_1/r_2))^{1/q}.$$

Also, if Γ is a not virtually abelian finitely generated group of polynomial growth that is quasi-isometric to G , we would have

$$c_1(B_n^\Gamma) \gtrsim_{\Gamma,q} (\log n)^{1/q}, \quad n \geq 2.$$

By the co-area formula it suffices to prove (26) when f is an indicator of a measurable set $A \subseteq G$,

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1 The endpoint question and scale profiles

Let G be a connected, simply connected nilpotent Lie group with Haar measure μ . A fixed bracket-generating horizontal space is equipped with a Euclidean norm, and $\nabla_H f$ denotes the corresponding left-invariant horizontal gradient. For $U \in \mathfrak{g}$, set

$$A_U^f(t) = \int_G |f(g \exp(tU)) - f(g)| d\mu(g), \quad t > 0. \quad (1)$$

If U is homogeneous of Carnot weight k , define its normalized logarithmic scale profile by

$$\Phi_U^\alpha[f](x) = e^{-\alpha x} A_U^f(e^x), \quad x \in \mathbb{R}, \quad \alpha = \frac{1}{k}. \quad (2)$$

For $1 < q < \infty$ and $1 \leq r \leq \infty$, let

$$\mathcal{B}_{\alpha;q,r}^U(f) = \|\Phi_U^\alpha[f]\|_{L^{q,r}(\mathbb{R})}. \quad (3)$$

Here $L^{q,r}$ is the usual Lorentz space with respect to Lebesgue measure in the logarithmic variable. Thus

$$\mathcal{B}_{\alpha;q,q}^U(f) = \left(\int_0^\infty \left(\frac{A_U^f(t)}{t^\alpha} \right)^q \frac{dt}{t} \right)^{1/q},$$

and the weak endpoint is

$$\mathcal{B}_{\alpha;q,\infty}^U(f) = \sup_{\lambda>0} \lambda |\{x \in \mathbb{R} : \Phi_U^\alpha[f](x) > \lambda\}|^{1/q}. \quad (4)$$

We write

$$\mathcal{H}_G(f) = \int_G |\nabla_H f(g)| d\mu(g). \quad (5)$$

If G has nilpotency step s , write $\mathfrak{g}_s = [\mathfrak{g}, [\mathfrak{g}, \dots, [\mathfrak{g}, \mathfrak{g}] \dots]]$ for its final nonzero lower-central layer. For $0 \neq v \in \mathfrak{g}_s$, Ryoo's endpoint question asks whether there is a finite q such that

$$\mathcal{B}_{1/s; q, q}^v(f) \lesssim_{G, v, q} \mathcal{H}_G(f) \quad \forall f \in C_c^\infty(G). \quad (6)$$

This is Question 38 in [8]; by coarea it is equivalent to the vertical perimeter formulation in Question 42 of that paper.

The elementary deterministic bound will be used repeatedly. Whenever a direction U admits horizontal paths from e to $\exp(tU)$ of length at most $C_U t^\alpha$ for every $t > 0$, differentiation along such a path and right invariance of Haar measure give

$$\|\Phi_U^\alpha[f]\|_{L^\infty(\mathbb{R})} \lesssim_{G, U} \mathcal{H}_G(f). \quad (7)$$

This applies to $U \in V_k$ in a Carnot group with $\alpha = 1/k$. It also applies, with the same exponent, to a fixed k -fold horizontal bracket in an arbitrary step-at-most- k nilpotent group, and to finite sums of such brackets when they are central. In particular it applies with $\alpha = 1/s$ to every final-layer direction used below. The difficulty is therefore not a bound at one scale, but summability over all scales.

Main conclusions of this note

The central new reduction is the following informal statement, proved precisely in sections 5 and 7:

A finite weak scale exponent is enough.

Indeed, weak L^r plus (7) gives strong L^q for every $q > r$. In step three, it is enough to prove this weak estimate for one direction on one group, the Cartan group.

2 The complete step-two theorem

We first verify and strengthen the supplied partial result. In this section G is any connected, simply connected, step-two nilpotent group, not necessarily Carnot graded. Let $W \subseteq \mathfrak{g}$ be a bracket-generating horizontal space with its fixed norm. Then $[\mathfrak{g}, \mathfrak{g}] = [W, W]$ is central. For $z \in [\mathfrak{g}, \mathfrak{g}]$, define

$$\kappa_W(z) = \inf \left\{ \sum_{j=1}^m \sqrt{|a_j| |b_j|} : z = \sum_{j=1}^m [a_j, b_j], \ a_j, b_j \in W \right\}. \quad (8)$$

Theorem 2.1 (Complete step-two endpoint theorem). *There is a universal constant $C < \infty$ such that every $f \in C_c^\infty(G)$ and $z \in [\mathfrak{g}, \mathfrak{g}]$ satisfy*

$$\left(\int_0^\infty \left(\frac{A_z^f(t)}{\sqrt{t}} \right)^4 \frac{dt}{t} \right)^{1/4} \leq C \kappa_W(z) \int_G |\nabla_W f| d\mu. \quad (9)$$

Consequently the same estimate holds for every strong exponent $q \geq 4$, with a constant depending on q . In particular, Ryoo's endpoint question is completely resolved in nilpotency step two.

Lemma 2.2 (Elementary Heisenberg transfer). *Let $a, b \in W$ and $z = [a, b] \neq 0$. Then*

$$\left(\int_0^\infty \left(\frac{A_z^f(t)}{\sqrt{t}} \right)^4 \frac{dt}{t} \right)^{1/4} \lesssim \sqrt{|a| |b|} \int_G |\nabla_W f| d\mu. \quad (10)$$

Proof. The vectors a, b, z span a three-dimensional Heisenberg Lie algebra. The connected subgroup

$$H = \exp(\text{span}\{a, b, z\})$$

is closed because the exponential map of a simply connected nilpotent group is a diffeomorphism and the displayed linear subspace is closed. Both G and H are unimodular. Weil disintegration therefore gives

$$\int_G \Psi(g) d\mu(g) = \int_{G/H} \int_H \Psi(xh) d\mu_H(h) d\bar{\mu}(x).$$

On each left coset, apply the sharp three-dimensional Heisenberg inequality of Naor and Young [5] to $h \mapsto f(xh)$. Minkowski's integral inequality in $L^4(dt/t)$ and Weil disintegration yield

$$\mathcal{B}_{1/2;4,4}^z(f) \lesssim \int_G (|af| + |bf|) d\mu.$$

Replace (a, b) by $(\lambda a, \lambda^{-1}b)$ and minimize $\lambda|a| + \lambda^{-1}|b|$ over $\lambda > 0$. This gives (10). $\square$

Proof of theorem 2.1. Choose a representation $z = \sum_{j=1}^m [a_j, b_j]$ and set $z_j = [a_j, b_j]$. Since all z_j are central, $\exp(tz) = \prod_j \exp(tz_j)$. Telescoping and right invariance of Haar measure give

$$A_z^f(t) \leq \sum_{j=1}^m A_{z_j}^f(t).$$

Minkowski and theorem 2.2 imply

$$\mathcal{B}_{1/2;4,4}^z(f) \lesssim \left(\sum_{j=1}^m \sqrt{|a_j||b_j|} \right) \int_G |\nabla_W f| d\mu.$$

Take the infimum over all representations to prove (9).

A commutator rectangle of side lengths r, u reaches $\exp(ru[a, b])$ with horizontal length $2r|a| + 2u|b|$. Optimizing under $ru = t$, summing over a representation of z , and then taking the infimum yields

$$\sup_{t>0} \frac{A_z^f(t)}{\sqrt{t}} \lesssim \kappa_W(z) \int_G |\nabla_W f| d\mu.$$

Interpolation of the scalar scale profile between L^4 and L^∞ proves all $q \geq 4$. $\square$

Remark 2.3. The exponent 4 is not always optimal. It is sharp for the three-dimensional Heisenberg group, whereas for $\mathbb{H}^{2k+1}$, $k \geq 2$, the sharp known exponent is 2; see [4, 5]. The point of theorem 2.1 is existence for every step-two group and every commutator direction.

3 Permanence principles

The reductions below use three elementary permanence facts. They are stated for the strong norm and for the finite weak norm. The weak case uses the equivalent Banach function norm, valid for $q > 1$,

$$\|F\|_{(q,\infty)} = \sup_{0 < |E| < \infty} |E|^{-1/q'} \int_E |F(x)| dx, \quad (11)$$

which is equivalent to the distribution-function expression in (4).

Lemma 3.1 (Coset transfer). *Let $H \leq G$ be a closed connected subgroup. Suppose an estimate in a direction $U \in \mathfrak{h}$, with either a strong L^q scale norm or a weak $L^{q,\infty}$ scale norm, holds on H with horizontal derivatives belonging to the ambient horizontal space. Then the same estimate holds on G , with a constant changed only by the fixed norm comparison.*

Proof. Apply the estimate to $h \mapsto f(xh)$ on each left coset. For a strong norm, integrate in G/H and use Minkowski. For a weak norm, use (11); its definition gives the corresponding Minkowski integral inequality directly. Finish with Weil disintegration. $\square$

Proposition 3.2 (Descent through nilpotent quotients). *Let $\pi : G \rightarrow \bar{G}$ be a surjective homomorphism between connected, simply connected nilpotent groups, with the quotient horizontal structure. If a strong or finite weak scale estimate holds on G in direction U , then it holds on $\bar{G}$ in direction $\pi_*U \neq 0$.*

Proof. It is enough to treat a quotient by a connected central subgroup, since every ideal in a finite-dimensional nilpotent Lie algebra admits a finite filtration with central successive quotients. Write the central quotient measure-theoretically as $G \cong \bar{G} \times \mathbb{R}^m$. Given $\bar{f} \in C_c^\infty(\bar{G})$, choose a smooth cutoff η_R on $\mathbb{R}^m$ with mass $m_R \rightarrow \infty$, and put $F_R(\bar{g}, z) = \bar{f}(\bar{g})\eta_R(z)$. After division by m_R , the horizontal energy converges to that of $\bar{f}$, while for every fixed scale the normalized translation integral converges to the corresponding quotient translation integral. Fatou's lemma proves the strong case. For the weak case, apply Fatou inside (11), then take the supremum over measurable scale sets. Iterate the central filtration. $\square$

Lemma 3.3 (Finite central sums). *Let $U_1, \dots, U_N$ be commuting directions. If a strong or finite weak scale estimate with the same homogeneity and exponent holds for each U_j , then it holds for $U_1 + \dots + U_N$.*

Proof. Central telescoping gives $A_{U_1+\dots+U_N}^f(t) \leq \sum_j A_{U_j}^f(t)$. Use the triangle inequality in the strong scale space or in the equivalent weak norm (11). $\square$

4 A universal Lorentz mixed-bracket theorem

The following theorem is the main propagation mechanism. It does not require W or $[X, W]$ to be central.

Lemma 4.1 (Universal anisotropic bracket word). *For every nilpotency bound $s \geq 2$, there are a finite number of coefficients and a word, depending only on s , such that for every step-at-most- s nilpotent group, every $X, W \in \mathfrak{g}$, and every $r, u > 0$,*

$$\exp(ru[X, W]) = \prod_{j=1}^{N(s)} \exp(c_j \lambda_j(r, u) \Sigma_j), \quad (12)$$

where $\Sigma_j \in \{X, W\}$, and $\lambda_j = r$ when $\Sigma_j = X$, while $\lambda_j = u$ when $\Sigma_j = W$.

Proof. Work in the free step- s nilpotent group on generators x, w . Let S be the set of finite products of elements of the one-parameter subgroups $\exp(\mathbb{R}x)$ and $\exp(\mathbb{R}w)$. It is a subgroup. By Chow–Rashevskii (equivalently, by the orbit theorem), the reachable set from the identity has nonempty interior because x, w Lie-generate the free algebra. Hence S is an open subgroup; the free group is connected, so S is the whole group. In particular $\exp([x, w])$ is a finite word in those two one-parameter subgroups. Apply the multigraded automorphism $x \mapsto rx$, $w \mapsto uw$, and then the universal homomorphism $x \mapsto X$, $w \mapsto W$. $\square$

Let $D_h(f) = \int_G |f(gh) - f(g)| d\mu(g)$. Unimodularity gives $D_{h_1 \dots h_N}(f) \leq \sum_j D_{h_j}(f)$. Thus theorem 4.1 implies

$$A_{[X,W]}^f(ru) \lesssim_s \sum_j A_X^f(c_j r) + \sum_j A_W^f(d_j u), \quad (13)$$

with fixed positive constants after replacing signs by absolute values.

Theorem 4.2 (Lorentz mixed-bracket propagation). *Let G have nilpotency step at most s , let $X, W \in \mathfrak{g}$, let $\alpha > 0$, $1 < q < \infty$, and $1 \leq r \leq \infty$. Put*

$$\beta = \frac{\alpha}{1 + \alpha}, \quad Q = (1 + \alpha)q, \quad R = (1 + \alpha)r, \quad (14)$$

with $R = \infty$ when $r = \infty$. Then

$$\mathcal{B}_{\beta;Q,R}^{[X,W]}(f) \lesssim_{s,\alpha,q,r} \left(\int_G |Xf| d\mu \right)^\beta (\mathcal{B}_{\alpha;q,r}^W(f))^{1-\beta}. \quad (15)$$

In particular:

$$\mathcal{B}_{\beta;Q,Q}^{[X,W]}(f) \lesssim \left(\int_G |Xf| d\mu \right)^\beta (\mathcal{B}_{\alpha;q,q}^W(f))^{1-\beta}. \quad (16)$$

$$\mathcal{B}_{\beta;Q,\infty}^{[X,W]}(f) \lesssim \left(\int_G |Xf| d\mu \right)^\beta (\mathcal{B}_{\alpha;q,\infty}^W(f))^{1-\beta}. \quad (17)$$

Proof. Set

$$a(r) = A_X^f(r), \quad b(u) = A_W^f(u), \quad c(t) = A_{[X,W]}^f(t), \quad M = \int_G |Xf| d\mu.$$

The path bound gives $a(r) \leq rM$. Define

$$B(y) = e^{-\alpha y} b(e^y), \quad C(x) = e^{-\beta x} c(e^x).$$

The finite word (13) gives, after absorbing its fixed dilations into a finite sum of translates $\tilde{B}$,

$$c(ru) \lesssim_s rM + u^\alpha \tilde{B}(\log u), \quad \|\tilde{B}\|_{L^{q,r}} \lesssim_s \|B\|_{L^{q,r}}. \quad (18)$$

Choose

$$u = \exp\left(\frac{x}{1 + \alpha} + z\right), \quad r = \exp\left(\frac{\alpha x}{1 + \alpha} - z\right).$$

Then $ru = e^x$, and (18) becomes

$$C(x) \lesssim_s M e^{-z} + e^{\alpha z} \tilde{B}\left(\frac{x}{1 + \alpha} + z\right). \quad (19)$$

Assume first $M > 0$. If $C(x) > \lambda$, choose $z = \log(c_s M / \lambda)$ with the constant adjusted so that the first term in (19) is at most $\lambda/2$. It follows that

$$\tilde{B}\left(\frac{x}{1 + \alpha} + z\right) \gtrsim_{s,\alpha} M^{-\alpha} \lambda^{1+\alpha}.$$

Writing $m_F(\lambda) = |\{F > \lambda\}|$, we obtain

$$m_C(\lambda) \leq (1 + \alpha) m_{\tilde{B}}(c_{s,\alpha} M^{-\alpha} \lambda^{1+\alpha}). \quad (20)$$

Equivalently, in terms of decreasing rearrangements,

$$C^*(t) \lesssim_{s,\alpha} M^\beta (\tilde{B}^*(c_\alpha t))^{1/(1+\alpha)}. \quad (21)$$

Multiply by $t^{1/Q}$, raise to the power $R = (1+\alpha)r$, and integrate dt/t . Since $Q = (1+\alpha)q$, this gives

$$\|C\|_{L^{Q,R}} \lesssim M^\beta \|\tilde{B}\|_{L^{q,r}}^{1/(1+\alpha)}.$$

Use (18). If $M = 0$, replace it by an arbitrary positive upper bound and let that bound decrease to zero. $\square$

Corollary 4.3 (Exact Carnot-weight recurrence). *Suppose $W_k \in V_k$ satisfies a strong or weak estimate with regularity $1/k$ and scale exponent q_k . For $a \in V_1$, set $W_{k+1} = [a, W_k]$. Then the same type of estimate holds for W_{k+1} , with*

$$q_{k+1} = \frac{k+1}{k} q_k. \quad (22)$$

Starting at weight two, this telescopes to

$$q_k = \frac{k}{2} q_2. \quad (23)$$

Proof. Apply theorem 4.2 with $\alpha = 1/k$. Then $\beta = 1/(k+1)$, and $Q = (k+1)q_k/k$. If both factors on the right are controlled by $\mathcal{H}_G(f)$, their weighted product is also controlled by $\mathcal{H}_G(f)$. $\square$

5 Why a finite weak exponent is enough

Lemma 5.1 (Weak-to-strong upgrade). *Let $0 < r < p < \infty$, and suppose $F \in L^{r,\infty}(\mathbb{R}) \cap L^\infty(\mathbb{R})$. Then*

$$\|F\|_{L^p} \leq \left(\frac{p}{p-r} \right)^{1/p} \|F\|_{L^{r,\infty}}^{r/p} \|F\|_{L^\infty}^{1-r/p}. \quad (24)$$

Proof. If $K = \|F\|_{L^{r,\infty}}$ and $M = \|F\|_{L^\infty}$, then $m_F(\lambda) \leq (K/\lambda)^r$ for $0 < \lambda \leq M$. The layer-cake formula gives

$$\|F\|_{L^p}^p = p \int_0^M \lambda^{p-1} m_F(\lambda) d\lambda \leq \frac{p}{p-r} K^r M^{p-r}.$$

$\square$

Corollary 5.2 (Finite weak endpoint suffices). *Let $\alpha > 0$, and suppose the deterministic profile bound $\|\Phi_U^\alpha[f]\|_\infty \lesssim \mathcal{H}_G(f)$ holds. If for some $1 < r < \infty$*

$$\mathcal{B}_{\alpha;r,\infty}^U(f) \lesssim \mathcal{H}_G(f), \quad (25)$$

then for every $p > r$,

$$\mathcal{B}_{\alpha;p,p}^U(f) \lesssim_p \mathcal{H}_G(f). \quad (26)$$

Proof. Combine theorem 5.1 with the assumed deterministic L^∞ profile bound. The applications below use $\alpha = 1/k$ for a weight- k direction, or $\alpha = 1/s$ for a final-layer direction in a step- s group. $\square$

This observation changes the remaining target substantially. One does not need to prove a sharp strong estimate at its endpoint. Any polynomial distribution bound for the bad scales is enough.

6 Universal finite models and rank-two seeds

6.1 A fixed-step free-model equivalence

Let $F_{m,s}$ denote the free step- s Carnot group on m horizontal generators.

Proposition 6.1 (Finite universal model at each step). *Fix $s \geq 2$. The following assertions are equivalent.*

(i) *Every connected, simply connected, step- s nilpotent group, equipped with any bracket-generating left-invariant horizontal structure, and every nonzero direction in its final lower-central layer satisfy Ryoo's inequality for some finite strong exponent.*

(ii) *Every basis vector of $V_s(F_{s,s})$ satisfies a finite weak-type estimate.*

If (ii) holds with a common weak exponent $r > 1$, then every step- s group in (i) satisfies the strong estimate for every $p > r$.

Proof. The implication (i) $\Rightarrow$ (ii) is immediate. Conversely, there are only finitely many chosen basis vectors. Increase the exponent past all of their weak exponents, use theorem 5.1, and then use finite central sums. This gives a common finite strong (hence weak) estimate for every vector in the finite-dimensional space $V_s(F_{s,s})$. Let W be the given horizontal space in an arbitrary step- s group. Since the image of W spans $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$, multilinearity of the bracket and the identity $[\mathfrak{g}_2, \mathfrak{g}_{s-1}] \subseteq \mathfrak{g}_{s+1} = 0$ show that the final layer $\mathfrak{g}_s$ is spanned by right-normed brackets

$$[a_s, [a_{s-1}, \dots, [a_2, a_1] \dots]], \quad a_j \in W.$$

Each such bracket is the quotient image of a top-layer vector in $F_{s,s}$. Use quotient descent, coset transfer, and central sums. Finally apply theorem 5.2. $\square$

6.2 The rank-two weak seed theorem

Let $F_{2,s}$ have free horizontal generators X, Y , and put

$$T = [X, Y] \in V_2(F_{2,s}). \quad (27)$$

For $s \geq 3$, this direction is noncentral.

Theorem 6.2 (One weak rank-two seed controls every group). *Fix $s \geq 2$. Suppose that for some $1 < q_2 < \infty$,*

$$\mathcal{B}_{1/2; q_2, \infty}^T(f) \lesssim_{s, q_2} \int_{F_{2,s}} (|Xf| + |Yf|) \quad \forall f \in C_c^\infty(F_{2,s}). \quad (28)$$

Then for every connected, simply connected, step- s nilpotent group G , every bracket-generating horizontal space W , every $v \in \mathfrak{g}_s$, and every $p > sq_2/2$,

$$\mathcal{B}_{1/s; p, p}^v(f) \lesssim_{G, W, v, p} \int_G |\nabla_W f| d\mu. \quad (29)$$

Before the weak-to-strong upgrade, the final direction satisfies weak exponent $sq_2/2$.

Proof. By the spanning argument in theorem 6.1, it is enough to treat a right-normed bracket

$$v = [a_s, [a_{s-1}, \dots, [a_3, [a_2, a_1]] \dots]], \quad a_j \in W.$$

Map the free generators X, Y to a_2, a_1 . The subgroup they generate is a nilpotent quotient of $F_{2,s}$. Thus (28) descends to the direction $[a_2, a_1]$, and then transfers from that subgroup to G . Apply the weak form (17) successively with $a_3, \dots, a_s$. By (23), the final weak exponent is $sq_2/2$. Sum the finitely many central right-normed terms and use theorem 5.2. $\square$

Remark 6.3. The natural hoped-for seed is $q_2 = 4$, inherited from the three-dimensional Heisenberg quotient. A weak lift of that one estimate through the central extensions $F_{2,s} \rightarrow F_{2,s-1}$ would give the universal strong conclusion for every $p > 2s$. The reverse lift through a central extension is precisely where the current argument stops.

7 The Cartan group is the complete step-three test

Let $C = F_{2,3}$ be the five-dimensional Cartan group, with Lie algebra

$$[X, Y] = T, \quad [X, T] = U, \quad [Y, T] = V, \quad (30)$$

where U, V are central. Its grading has dimensions $(2, 1, 2)$.

7.1 Polarization and symmetry

For $P(a, b) = [a, [a, b]]$, the identity

$$3[a, [b, c]] = P(a + b, c) - P(a, c) - P(b, c) - P(a + c, b) + P(a, b) + P(c, b) \quad (31)$$

shows that every triple bracket is a linear combination of brackets involving only two horizontal vectors. Each $P(a, b)$ is the image of $U = [X, [X, Y]]$ under the homomorphism $C \rightarrow G$ sending $X \mapsto a, Y \mapsto b$.

Moreover, $\mathrm{GL}_2(\mathbb{R})$ acts by graded automorphisms on C . On $V_3(C) = \mathrm{span}\{U, V\}$, this action is the standard two-dimensional action twisted by the determinant, and is transitive on nonzero vectors. Thus all nonzero top-layer directions of C are equivalent.

Theorem 7.1 (Exact weak Cartan reduction). *The following are equivalent.*

- (i) *Ryoo's endpoint conjecture has a positive answer for every connected, simply connected, step-three nilpotent group with every bracket-generating left-invariant horizontal structure.*
- (ii) *For some $1 < r < \infty$, every $f \in C_c^\infty(C)$ satisfies*

$$\boxed{\mathcal{B}_{1/3;r,\infty}^U(f) \lesssim_r \int_C (|Xf| + |Yf|).} \quad (32)$$

If (32) holds, then every group and horizontal structure in (i) satisfies the strong inequality for every exponent $p > r$.

Proof. The implication (i) $\Rightarrow$ (ii) follows by applying (i) to C , since a strong estimate implies a weak estimate. Conversely, the graded automorphism action gives the same weak estimate for every top direction in C . Let W be a bracket-generating horizontal space in an arbitrary step-three nilpotent group. Its final layer is spanned by $[a, [b, c]]$ with $a, b, c \in W$. The polarization identity (31) writes each such vector as a finite sum of terms $P(u, w) = [u, [u, w]]$, each of which is a quotient image of $U = [X, [X, Y]]$. Quotient descent, coset transfer, and central sums therefore transfer the weak estimate to every final-layer direction. Apply theorem 5.2. $\square$

Thus the whole step-three problem is not merely reduced to Cartan in a heuristic sense: it is equivalent to one explicit weak distribution estimate on that group.

7.2 A stronger but reusable noncentral seed

A particularly useful sufficient statement concerns the second-layer direction T .

Corollary 7.2 (Weak Cartan seed in the noncentral direction). *Suppose that for some $1 < q_2 < \infty$,*

$$\mathcal{B}_{1/2; q_2, \infty}^T(f) \lesssim \int_C (|Xf| + |Yf|). \quad (33)$$

Then U and V satisfy weak exponent $3q_2/2$, and every step-three nilpotent group with a bracket-generating horizontal structure satisfies the strong endpoint inequality for every $p > 3q_2/2$. In particular, the natural quotient-Heisenberg value $q_2 = 4$ would give every strong exponent $p > 6$.

Proof. Apply (17) to $U = [X, T]$ and $V = [Y, T]$ with $\alpha = 1/2$. Then $\beta = 1/3$ and $Q = 3q_2/2$. Finish with theorems 5.2 and 7.1. $\square$

Proposition 7.3 (A concrete absorption criterion). *Suppose the following twisted Heisenberg lifting estimate holds on C :*

$$\mathcal{B}_{1/2; 4, 4}^T(f) \lesssim \mathcal{H}_C(f) + \mathcal{B}_{1/3; 6, 6}^U(f) + \mathcal{B}_{1/3; 6, 6}^V(f). \quad (34)$$

Then

$$\mathcal{B}_{1/2; 4, 4}^T(f) + \mathcal{B}_{1/3; 6, 6}^U(f) + \mathcal{B}_{1/3; 6, 6}^V(f) \lesssim \mathcal{H}_C(f). \quad (35)$$

Consequently every step-three nilpotent group with a bracket-generating horizontal structure satisfies the strong endpoint inequality for every $q \geq 6$.

Proof. The strong mixed-bracket theorem gives

$$\mathcal{B}_{1/3; 6, 6}^U(f) + \mathcal{B}_{1/3; 6, 6}^V(f) \lesssim \mathcal{H}_C(f)^{1/3} \mathcal{B}_{1/2; 4, 4}^T(f)^{2/3}.$$

Insert this into (34) and use Young's inequality with a sufficiently small coefficient to absorb the T -term. The estimates for U, V then follow from the displayed mixed bound. Apply theorem 7.1 and the deterministic profile bound to obtain all $q \geq 6$. $\square$

Thus (34) is a second, stronger formulation of the missing analytic step. Unlike the weak target (32), it is scale-critical and has a specified exponent; its advantage is that it is directly compatible with absorption.

8 Unconditional higher-step families

The propagation theorem already proves nontrivial examples in every step.

Theorem 8.1 (Step-three groups of type $\star$). *Let G be a step-three Carnot group of type $\star$ in the sense of Marchi [3]. Thus V_1 has a basis $X_1, \dots, X_m$ satisfying*

$$[X_j, [X_j, X_i]] = 0 \quad \forall i, j. \quad (36)$$

Then every $v \in V_3$ satisfies

$$\mathcal{B}_{1/3;6,6}^v(f) \lesssim_{G,v} \mathcal{H}_G(f). \quad (37)$$

Hence every exponent $q \geq 6$ is admissible.

Proof. For each pair i, j , the relations (36) imply that $W_{ij} = [X_j, X_i]$ commutes with both X_i and X_j . If $W_{ij} \neq 0$, these three vectors generate a three-dimensional Heisenberg subgroup. The Naor–Young theorem and coset transfer therefore give the strong $(1/2, 4)$ estimate for W_{ij} . Since $V_3 = [V_1, V_2]$, every $v \in V_3$ is a finite sum of vectors $[X_k, W_{ij}]$. Apply the strong mixed-bracket theorem with $\alpha = 1/2$ to obtain exponent 6 for each term, and then use central telescoping. $\square$

Theorem 8.2 (Upper unitriangular groups). *Let $UT_{s+1}(\mathbb{R})$ have its standard Carnot grading and horizontal generators $X_i = E_{i,i+1}$, $1 \leq i \leq s$. For the top direction $Z = E_{1,s+1}$,*

$$\mathcal{B}_{1/s;2s,2s}^Z(f) \lesssim_s \mathcal{H}_{UT_{s+1}}(f). \quad (38)$$

Hence every exponent $q \geq 2s$ is admissible for this direction.

Proof. Set $W_2 = [X_{s-1}, X_s] = E_{s-1,s+1}$. The subgroup generated by X_{s-1}, X_s is three-dimensional Heisenberg, so the Naor–Young theorem and coset transfer give strong exponent $q_2 = 4$ for W_2 . Recursively define

$$W_k = [X_{s-k+1}, W_{k-1}] = E_{s-k+1,s+1}, \quad 3 \leq k \leq s.$$

Apply theorem 4.3; $q_k = (k/2)4 = 2k$. At $k = s$, $W_s = Z$. $\square$

This family gives a positive endpoint theorem in every nilpotency step without assuming any unproved seed.

9 Why the Cartan estimate does not follow by the obvious lifts

The quotient $C/\exp(\text{span}\{U, V\})$ is the three-dimensional Heisenberg group, where the T -estimate with exponent 4 is known. It is tempting to lift that estimate back to C , but the two simplest routes fail for structural reasons.

9.1 Failure of plain quotient lifting

A section of the central quotient turns quotient horizontal derivatives into ambient horizontal derivatives plus top-layer drift. In any fixed trivialization the two lifted horizontal fields cannot both be made drift-free, because the connection has curvature. The resulting U - and V -derivatives are not controlled in L^1 by $|Xf| + |Yf|$. Averaging over larger and larger central fibers proves descent from an extension to its quotient, as in theorem 3.2, but gives no reverse estimate.

9.2 Failure of plain Heisenberg slicing

The subgroup $\exp(\text{span}\{X, T, U\})$ is a genuine Heisenberg group. Slicing by its left cosets and applying Naor–Young controls top translation in U by the derivatives Xf and Tf . The latter is not an ambient horizontal derivative. The analogous subgroup $\exp(\text{span}\{Y, T, V\})$ has the same defect.

Proposition 9.1 (No first-derivative substitute for compensation). *There is no constant C_0 such that*

$$\|Tf\|_{L^1(C)} \leq C_0(\|Xf\|_{L^1(C)} + \|Yf\|_{L^1(C)}) \quad \forall f \in C_c^\infty(C). \quad (39)$$

Proof. Use polynomial coordinates (x, y, a, b, c) in which

$$Y = \partial_y, \quad X = \partial_x - y\partial_a - \frac{y^2}{2}\partial_b - xy\partial_c, \quad T = \partial_a + y\partial_b + x\partial_c.$$

Choose nonzero smooth compactly supported one-variable cutoffs and set

$$f_\varepsilon(x, y, a, b, c) = \chi_1(x)\eta(y/\varepsilon)\varphi(a/\varepsilon^2)\chi_2(b)\chi_3(c).$$

Choose χ_2, χ_3 equal to one on a fixed neighborhood and choose $\varphi' \not\equiv 0$. Direct scaling gives

$$\|Yf_\varepsilon\|_1 = O(\varepsilon^2), \quad \|Xf_\varepsilon\|_1 = O(\varepsilon^2), \quad \|Tf_\varepsilon\|_1 \gtrsim \varepsilon.$$

For the last inequality, restrict to the region where $\chi_2 = \chi_3 = 1$; there the ∂_a -term has size $\asymp \varepsilon$ in L^1 , while the remaining terms are of smaller order. Letting $\varepsilon \downarrow 0$ contradicts (39). $\square$

Therefore a successful argument must use compensation or multiscale cancellation, not a pointwise derivative comparison.

9.3 Exact commutator geometry

The Baker–Campbell–Hausdorff formula gives, up to harmless sign conventions,

$$\exp(aX)\exp(bY)\exp(-aX)\exp(-bY) = \exp\left(abT + \frac{a^2b}{2}U + \frac{ab^2}{2}V\right). \quad (40)$$

The top-layer errors have exactly the correct ambient homogeneity: when $a \asymp b \asymp \sqrt{t}$, they have size $t^{3/2}$, whose Carnot distance is $t^{1/2}$. Thus the obstruction is not a mismatch of powers. It is the need to sum the horizontal cost over many scales without charging the entire perimeter at each scale.

10 Kinetic and Rumin formulations of the missing estimate

Continue in the polynomial coordinates used in the proof of theorem 9.1:

$$Y = \partial_y, \quad X = \partial_x - y\partial_a - \frac{y^2}{2}\partial_b - xy\partial_c. \quad (41)$$

For an intrinsic graph $y = \psi(x, a, b, c)$, the horizontal slope is governed by

$$\partial_x\psi - \partial_a\frac{\psi^2}{2} - \partial_b\frac{\psi^3}{6} - x\partial_c\frac{\psi^2}{2}.$$

Introducing a kinetic parameter ξ , define

$$L_\xi = \partial_x - \xi \partial_a - \frac{\xi^2}{2} \partial_b - x \xi \partial_c. \quad (42)$$

The exact commutator identities are

$$[L_\xi, \partial_\xi] = \partial_a + \xi \partial_b + x \partial_c, \quad (43)$$

$$[L_\xi, [L_\xi, \partial_\xi]] = \partial_c, \quad (44)$$

$$[\partial_\xi, [L_\xi, \partial_\xi]] = \partial_b, \quad (45)$$

$$[L_\xi, L_\eta] = (\xi - \eta) \partial_c. \quad (46)$$

These identities show that the missing top-layer estimate is a critical kinetic transfer-of-regularity statement. The closest available trajectory-based theorem reaches the critical regularity in a Besov class with fine index ∞ [6]; the desired result asks for a finite distribution exponent, or equivalently a polynomial bound on the number of bad logarithmic scales.

There is also a Rumin-complex formulation. The horizontal gradient $d_c f = (Xf, Yf)$ is not an arbitrary L^1 vector field: it satisfies the compatibility relation $d_c(d_c f) = 0$. A successful proof should exploit this cancellation to solve a multiscale horizontal divergence equation. For example, standard L^6 – $L^{6/5}$, L^1 – L^∞ , and Hahn–Banach duality reduces the strong top-layer estimate at exponent 6 to the following construction. For every family ϕ_t satisfying

$$\|\phi\|_{L^{6/5}((0,\infty), dt/t; L^\infty(C))} \leq 1,$$

one seeks a horizontal vector field Ψ with

$$\operatorname{div}_H \Psi = \int_0^\infty \frac{\phi_t(\cdot \exp(-tU)) - \phi_t}{t^{1/3}} \frac{dt}{t}, \quad \|\Psi\|_\infty \lesssim 1. \quad (47)$$

Indeed, integration by parts would then bound the dual pairing by $\mathcal{H}_C(f)$. Solving each scale separately causes a logarithmic divergence; the desired theorem is exactly the missing cancellation across scales.

Recent endpoint Poincaré and Hodge estimates on the Cartan group [1], critical kinetic Besov estimates [6], and nonlocal Sobolev–BV characterizations on Carnot groups [7] are highly relevant to this formulation. The Cartan estimates are spatial, the kinetic theorem reaches the critical regularity with fine index ∞ , and none of the three papers states the directional finite logarithmic-scale summability in (32).

11 Counterexample audit and literature status

Ryoo’s version of record, accepted on March 22, 2026 and available online in April 2026 (with issue publication listed for August 1, 2026), still states the endpoint problem as Questions 38 and 42 and Conjecture 43 [8]. The sharp Heisenberg inputs are the $q = 2$ result in dimensions at least five [4] and the sharp $q = 4$ result in dimension three [5]. The theorem that general nonabelian nilpotent groups do not embed bi-Lipschitzly into L^1 is qualitative and does not provide a finite scale-summability exponent [2].

A literature search through June 19, 2026 did not locate a later proof or counterexample for Ryoo’s finite-exponent endpoint conjecture. Several plausible counterexample sources were examined:

- pathological constant-normal sets in the Engel and Cartan groups;
- translated homogeneous cones in the Cartan group;
- critical kinetic examples of De Lellis–Westdickenberg type;
- rough intrinsic graphs in higher-step groups.

None of these produced a verified divergence of every finite scale norm. In the explicit Cartan cone models, the top-layer translation modulus appears at least as good as the critical homogeneity, so they do not presently furnish a counterexample. This negative search result is not a proof that no counterexample exists.

12 Final formulation of the remaining problem

After all reductions, the first unresolved universal statement can be written without any auxiliary group theory.

Conjecture 12.1 (Weak Cartan endpoint). *There exists $1 < r < \infty$ such that every $f \in C_c^\infty(C)$ satisfies*

$$\sup_{\lambda > 0} \lambda \left| \left\{ x \in \mathbb{R} : e^{-x/3} \int_C |f(g \exp(e^x U)) - f(g)| \, dg > \lambda \right\} \right|^{1/r} \lesssim_r \int_C (|Xf| + |Yf|).$$

By theorem 7.1, this conjecture is equivalent to a positive resolution of the endpoint problem for every step-three nilpotent group with every bracket-generating left-invariant horizontal structure. By theorem 5.2, it immediately yields strong estimates at every larger exponent. Alternatively, the stronger noncentral weak seed (33) with $q_2 = 4$ would imply every strong step-three exponent $p > 6$, and a weak rank-two seed on $F_{2,s}$ would propagate to arbitrary step by theorem 6.2.

The final attempt therefore does not close the conjecture, but it reduces the first hard case to a strictly weaker and more concrete target than the original strong inequality: a polynomial distribution estimate for the bad scales of one Cartan translation profile.

References

- [1] A. Baldi and F. Rosa. L^p -Hodge decomposition and global integral estimates on the Cartan group. arXiv:2604.15206, 2026.
- [2] S. Eriksson-Bique, C. Gartland, E. Le Donne, L. Naples, and S. Nicolussi Golo. Nilpotent groups and bi-Lipschitz embeddings into L^1 . *International Mathematics Research Notices*, 2023(12):10759–10797, 2023.
- [3] M. Marchi. Regularity of sets with constant intrinsic normal in a class of Carnot groups. *Annales de l’Institut Fourier*, 64(2):429–455, 2014.
- [4] A. Naor and R. Young. Vertical perimeter versus horizontal perimeter. *Annals of Mathematics*, 188(1):171–279, 2018.
- [5] A. Naor and R. Young. Foliated corona decompositions. *Acta Mathematica*, 229(1):55–200, 2022.

- [6] L. Niebel. Transfer of regularity by kinetic mollification along critical trajectories. arXiv:2605.13582v2, 2026.
- [7] F. Serra Cassano and K. Zambanini. Characterizations of Sobolev and BV functions on Carnot groups. arXiv:2603.28996, 2026.
- [8] S.-Y. Ryoo. Quantitative nonembeddability of groups of polynomial growth into uniformly convex spaces. *Journal of Functional Analysis*, article 111491, 2026. DOI: 10.1016/j.jfa.2026.111491; arXiv:2207.11305v2.